# GroupBuy Dubai: Data Analytics for Business Decisions and Automation

Individual Data Analytics Assignment | Syamantak Dey

**Draft for review.** Check each result against the dashboard, rewrite the yellow passages in your own words, and delete this note before submitting.

## Startup and business objective

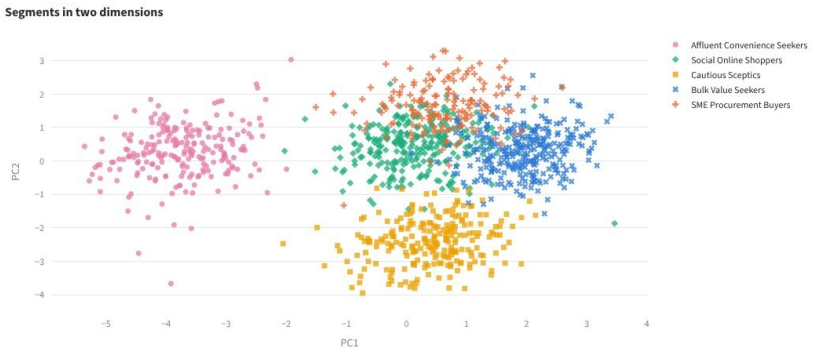
GroupBuy is a demand-first buying platform for Dubai. Buyers register what they want, suppliers bid against the pooled quantity, and buyers approve the final delivered price before paying. GroupBuy earns a commission on fulfilled orders, assumed in the deck at 7%. The deck names its largest early risks as trust and unit economics, so this project asks: whom should GroupBuy target first, what should it pool together, how large will orders be, and which of these decisions can be automated?

## Data

The analysis uses a mock survey of 1,215 rows. Cleaning removed 15 duplicate submissions, leaving 1,200 respondents, and corrected text-formatted spend, impossible values such as an age of 120, and inconsistent labels. Because the deck contains no transactions, I generated a synthetic Year 1 of 120 campaigns and 4,449 buyer commitments, using the deck's commission, variable cost and fixed cost. Survey answers drive the simulation: for example, a buyer is more likely to pay when the net discount clears their stated minimum. None of this data is real market evidence.

## Clustering: who are the buyers?

K-Means was used because no segment labels exist. Thirteen behavioural variables were scaled, with intent answers left out. I chose five clusters: this is the elbow and a local silhouette peak. Two clusters scored higher but were too coarse to act on. A silhouette of 0.237 shows that the segments overlap. Bulk Value Seekers have the highest high-intent share (59%), while Affluent Convenience Seekers have 4% and need a 25% average discount. GroupBuy should launch with the first group and avoid spending on the second. New sign-ups can be assigned automatically to a segment and its message flow.




**Segment profile and action**

| Segment | Customers | Median spend AED | Min discount % | Max wait days | Median order AED | High intent |
|---|---|---|---|---|---|---|
| Bulk Value Seekers | 296 | 2650 | 12.1 | 6.9 | 685 | 59% |
| SME Procurement Buyers | 165 | 7580 | 9.5 | 9.8 | 1720 | 45% |
| Social Online Shoppers | 288 | 1580 | 15.0 | 5.0 | 350 | 22% |
| Cautious Sceptics | 231 | 1800 | 22.0 | 4.1 | 260 | 4% |
| Affluent Convenience Seekers | 220 | 4220 | 25.0 | 2.1 | 590 | 4% |

- **Bulk Value Seekers:** Core launch audience: staples and household campaigns with consolidated delivery.
- **SME Procurement Buyers:** Prioritise for high-value office and bulk campaigns; offer bid comparison and VAT invoices.
- **Social Online Shoppers:** Recruit through referral rewards and WhatsApp groups; run small-basket, short campaigns.
- **Cautious Sceptics:** Do not spend paid acquisition; nurture with verified-savings proof and refund guarantees.

*Figure 1. Dashboard: K-Means segments and profiles*

**[Rewrite in your own words]** What surprised you about the segments, and which one would you pursue first?

## Classification: which leads come first?

The target is whether a respondent is in the High conversion segment (27.6% of respondents). The conversion score is almost entirely a weighted sum of three intent answers, so these answers, the score and free-text comments were excluded to prevent leakage. Gender and nationality were also excluded so they cannot influence outreach priority. Random Forest was selected using cross-validated F1 on training data.

| Classifier (test set) | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| Always majority | 72.3% | 0% | 0% | 0.000 |
| Logistic Regression | 81.3% | 63.1% | 78.3% | 0.699 |
| Decision Tree | 75.7% | 54.3% | 75.9% | 0.633 |
| Random Forest (selected) | 77.7% | 56.6% | 83.1% | 0.673 |

On 300 test leads the confusion matrix gives 69 true positives (high-intent leads correctly called), 164 true negatives (correctly left in low-cost nurture), 53 false positives (wasted calls) and 14 false negatives (missed buyers). High accuracy would mean little here, since always predicting "not high" reaches 72.3%. During cold start a missed buyer costs more than a wasted message, so recall of 83.1% matters most. Logistic Regression did better on this one test set, so the choice between the two is not clear-cut. Sign-ups can be scored automatically and sent either to a founder call list or a WhatsApp nurture flow.

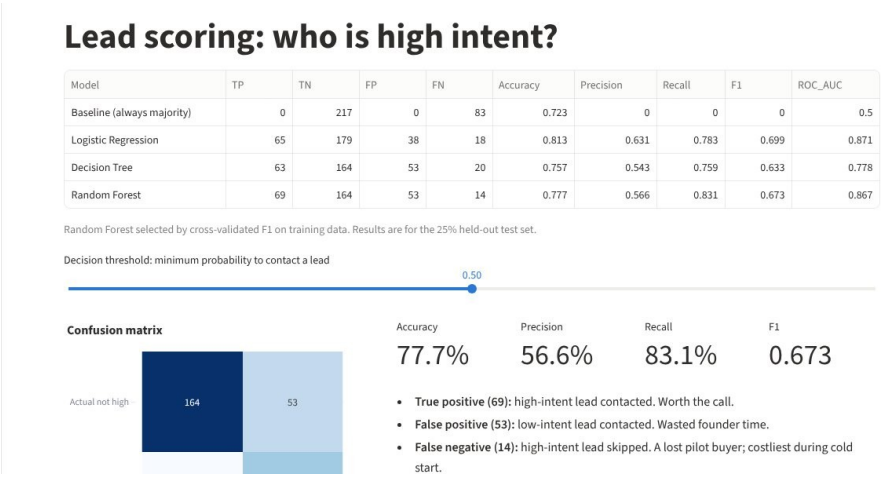



*Figure 2. Dashboard: classification comparison and confusion matrix*

**[Rewrite in your own words]** Explain in your own business terms why false negatives or false positives matter more for GroupBuy.

## Association rules: what belongs together?

Apriori treated each respondent's selected categories and features as a basket. Baby and groceries interest strongly implies fresh produce and meat (confidence 87%, lift 2.64, support 11%). Building pooling with WhatsApp updates implies consolidated delivery (confidence 86%, lift 2.59). This suggests family bundles on a shared delivery run and one packaged "building campaign". The rules could automatically suggest a related open campaign when a buyer commits. They show overlapping stated interest, not purchases or causation.

## Regression: how large will orders be?

Expected order value was predicted from 17 variables. Random Forest (MAE AED 225, MSE 117,008, RMSE AED 342, $R^2$ 0.71) beat Multiple Linear Regression ($R^2$ 0.47) and a mean baseline. Eligible monthly spend is the leading driver. Errors differ sharply by buyer type: mean error is AED 164 for individuals but AED 609 for SMEs, so the two should be forecast separately. Predicted value multiplied by expected paying buyers can automatically suggest a campaign target quantity.

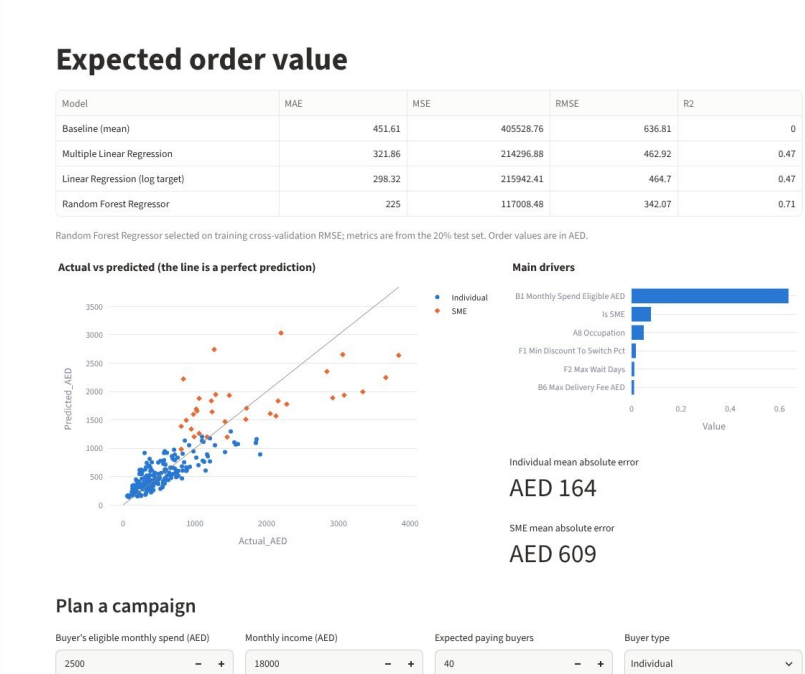



*Figure 3. Dashboard: actual versus predicted order value and campaign planner*

## Business performance and recommendations

The simulation produced AED 1.86M GMV, AED 130,331 commission and a loss of AED 86,906 after fixed costs. Break-even needs about AED 300,000 of monthly GMV, reached only in November. This exposes a tension in the deck: break-even of about 15 orders at AED 20,000 implies AED 3.6M of annual GMV, whereas its Year 1 forecast is AED 1.2M.

I recommend launching with Bulk Value Seekers in staples and household categories, running SMEs as a separate track, ensuring net discounts approach the 16% median minimum (simulated offers average 13.1%), leading with the refund-if-campaign-fails guarantee, and revisiting the Year 1 financial plan.

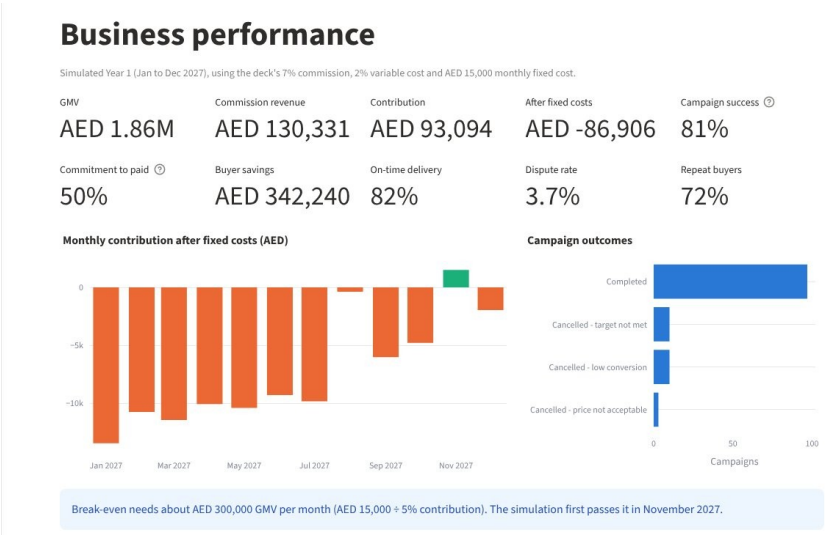



*Figure 4. Dashboard: simulated Year 1 performance*

## Limitations and learning

The models learn patterns that exist in mock data, so good scores do not prove real-world accuracy. Lead scoring predicts stated intent rather than payment. The next step is to retrain on actual pilot outcomes, in line with the deck's 90-day validation plan.

**[Rewrite in your own words]** What did you personally learn? Cover at least one thing about data leakage or metric choice, and one about turning a model output into a business decision.